Thierno Ibrahima Barry

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Presentation

As a Software Engineering graduate from Concordia University with over four years of professional experience, I bring a unique combination of academic rigor and proven industry expertise. My career has spanned multiple roles across diverse sectors—including web development, Internet of Things (IoT), cybersecurity, and robotics—demonstrating adaptability and a strong capacity to thrive in dynamic environments. This polyvalent background has equipped me with the ability to tackle complex challenges, bridge disciplines, and deliver impactful solutions across different technological landscapes.

Education

Concordia University,

Gina Cody School of Eng. And Computer Science — Montreal, QC

Bachelor of Engineering, Software Engineering (GPA: 3.43) — *Dec. 2019 – Apr 2024*

Collège De Maisonneuve — Montreal, QC

D.E.C, Natural Science — *Sep. 2016 – May 2018*

Experience

Intermediate Software Engineer — Aug 2024 – Current

MDA Space (JavaScript, React, Django, Spring, Java, SQL, Bash, Linux) — *Longueuil, QC*

- Utilized React and Django Rest Framework to build from scratch a Financial Update System (FUS) responsible for providing an intuitive, efficient, and secured system for processing invoices and payments.
- Developed core functionalities of the company satellite picture acquisitions with the Java Spring framework.
- Built a RESTful API in Django Rest Framework for notifying the clients of upcoming maintenance windows.
- Implemented, managed and deployed many Docker Containers of our different applications across the company servers, including monitoring and system-admin tasks.
- Performed a major migration of many legacy systems, upgrading them from Java-8 to Java-17 in a mission to decrease their vulnerabilities and increase their maintainability.

Software Developer — May 2022 – Sep 2023

Kinova Robotics (JavaScript, React, SQL, C++, Node.js, MQTT, RPC) — *Boisbriand, QC*

- Utilized React to build and optimize interactive user interfaces, ensuring a responsive and seamless user experience.
- Leveraged Node.js for server-side development to handle RESTful APIs and implement backend services.
- Integrated RPC (Remote Procedure Call) mechanisms to enable efficient communication between distributed components.
- Developed comprehensive Python scripts to automate testing processes and validate the functionality of robotic arms.
- Developed and optimized firmware code in C++ to facilitate communication between sensors and actuators, enhancing the robotic arm's performance and responsiveness.

Software Engineer — Sep. 2021 – Dec 2021

Hydro-Quebec (Python, Java, SpringBoot, API, SQL) — *Montreal, QC*

- Utilized the Checkpoint API to enhance firewall security, automating the management and configuration of firewall rules and policies.
- Created sophisticated Python scripts to automate the extraction, transformation, and loading (ETL) of network data from various sources, enhancing data processing efficiency.
- Utilized Java Spring Boot's RESTful API capabilities to expose endpoints for data retrieval and manipulation, enabling seamless integration with other systems and applications.

Software Developer May 2021 – Sep 2021

PARADIGM (C#, React, SQL, .NET) *Repentigny, QC*

- Utilized React to build and enhance dynamic and responsive user interfaces, ensuring an intuitive and engaging user experience for the company web platform.
- Employed C# and the .NET framework to develop robust backend services and APIs, facilitating seamless communication between the frontend and backend components.

Enterprise Architect May 2019 – Sep 2019

National Bank of Canada (Microsoft Suit Office 365) *Montreal, QC*

- Categorization and detailed mapping of the different software used by the company.
- Investigated and reported about the potential acquisition of a new company-wide tool.

Projects

Spark-Solution | *Vite, React Java (Spring), Maven, Rust, Redis, Postgres* Oct 2025 – Present

- A smart parking solution providing real-time spot availability and navigation via sensors.
- Designed the system architecture, the app frontend/backend alongside the sensors firmware.

CSSE Accounting System | *React, Python, Django, SQL* June 2025 – Present

- An accounting system for the family school establishment in Guinea.
- Provide a beautiful and efficient interface for registering and processing the student's tuition fees.
- Will be used as a base to expend into a complete School management application.

Education-MindA.I.lytics | *Python, Pytorch, Flask, ReactJs, Docker, RTL, Cypress* Dec 2024 – Present

- A Deep Learning Convolutional Neural Network (CNN) built on Pytorch.
- Trained the A.I on a massive set of facial emotion.
- Cleaned and labeled data.

IoT-Flood-Drain | *C++, Arduino, HTML, JavaScript, Node.js, Git* Sep 2023 – Present

- Built an electrical system composed of an Arduino, multiples sensors and a water pump.
- Developed a Water Diagnostic Web Server using Node.js.
- The system allows to drain a flood automatically once a certain water level is detected.

Technical Skills

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, R, PHP, SHELL, C#, Bash
Frameworks: React, Node.js, Flask, JUnit, WordPress, Material-UI, FastAPI, Angular, Vue, Laravel, Express, .NET, Spring, PyScript, RTL, Django Rest Framework
Developer Tools: Git, Docker, TravisCI, Google Cloud Platform, VS Code, Visual Studio, Eclipse, AWS, Azure, PowerShell, SSH, Linux, Jenkins, Postman, Cypress
Certificate: Docker, Git, AWS, Azure, Agile, Claude Code